

TOM G. HARDY

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An **award-winning** 4th year physics and astronomy MPhys Student at **Durham University** with **three summers'** academic research experience (including at **Cambridge** and **Oxford**); a particular interest in **galaxy evolution** and **astronomy** in the **large-survey** era. Experience in astronomy from **LIGO** to **MeerKAT**, and past projects have covered from ML statistical inference all the way to exoplanetary detection.

EDUCATION

2022-2026: Durham University – MPhys Physics & Astronomy – Predicted High 1st Class Honours

Awarded Florence Nightingale (AstroLab Report) & first year Laboratory (**highest mark in the year**) **transcript awards**; co-chair of the Physics staff-student committee; lead student representative to the **Board of Studies**. Overseen **curriculum changes** in the physics department. 2nd year research-led-lab in **top-10** marks awarded, **96%** 2nd-year computational physics sub-module score. **82% score** in eclipsing binaries 3rd year 17-page report (**BSc report equivalent**), **highest in Durham Astrolab**. Four reports shortlisted for transcript awards.

Creative director of largest DU music society, and Composer for the Jazz Orchestra (first year); Conductor/Composer for the Jazz Orchestra (second year); Musical Director for the Big Band (third & fourth year). Raised over **£3,000** for charity, led outreach in low-income Northeast communities **and awarded highest Full Palatinate award**. Managed and awarded over **£8,000** in arts grants.

2015-2022: John Hampden Grammar School – 5 A-Levels (5A*s), 11 GCSEs (all grade 9)

Nationally leading A-level and GCSE results, with maximum possible grades scored. Deputy Head Boy, Director of AV, Mock Trials (law) four-time regional finalist. First President of the Euler-Erdos Mathematical Society. Awarded over 30 academic excellence awards and the Carl Mullen Trophy.

RESEARCH EMPLOYMENT AND PROJECTS

One second-author paper in the Journal of Physics Education. **Two** papers in prep; one first author on SBI for model-misspecification tests on LIGO, one co-author with Oxford MIGHTEE-HI group. **Four** research seminars given across Oxford and Cambridge, including **an invited talk** at the IoA's 30-year anniversary event.

~ Summer Projects ~

Summer 2025 – Institute of Astronomy, Cambridge University

'Simulation Based Inference to detect Gravitational Wave Mismodelling'

Supervisor: Dr. James Alvey

Machine learning simulation-based inference to build frequentist-style mismodelling detection tools for gravitational wave astronomy. The project used differentiable waveform simulators and LIGO noise to train networks capable of detecting physically-motivated phase and amplitude distortions in raw GW data. Involved significant GPU programming and neural architecture; additionally international collaboration with MPA Garching. Organised and chaired a weekly interns' group meeting and was **invited back as a speaker for the IoA's 30-year anniversary event**.

Summer 2024 – Astrophysics, Department of Physics, Oxford University

'Source Extraction for Radio-Detected HI Galaxies'

Supervisors: Dr Catherine Hale, Prof Matt Jarvis, Dr Anastasia Ponomareva

Research Intern in Hintze centre for Astrophysical Surveys/Galaxies sub-group. I built PHiSEP, a neutral hydrogen source extraction pipeline using Aaron Robotham's ProFound. The project was initially designed to evaluate the effectivity of segmentation extractors on HI-data-cubes, and led to the development of an MPI-enabled HPC (Oxford Glamdring) pipeline (using python and R) for MIGHTEE-HI/COSMOS fields, tested on simulated SKA data. Attended the Breakthrough Discuss and Steven Balbus conferences. Project funded by the UNIQ+ scholarship.

Summer 2023 – PhES, Department of Physics, Durham University

'The Impact of AI in Physics Education: A Comprehensive Review from GCSE to University Levels'

Supervisor: Dr Will Yeadon

Research intern in the Durham PhES sub-group. A comprehensive statistical investigation into the power of early LLMs (and prompting engineering) to solve academic exams with the intent of creating deep learning-ready database for LLM training. The project involved use of python regex infrastructure to transcribe large PDF data (for analysis); and early GPT-API usage with prompt engineering. This project also led to a 2024 Laidlaw Scholarship project on deep LLM learning using this data. Work published in Journal of Physics Education (**cited by 100+**).

September 2025-May 2026 – CEA, Department of Physics, Durham University (MPhys Project)

‘Identifying Extreme Emission Line Galaxies (EELGs) from VLT MUSE observations’

Supervisor: Prof Alastair Edge

MUSE IFU spectroscopy to detect strongly lensed images of low-Z EELGs in nearby massive field clusters, targeted on the OIII line. The project aims to infer an aurora-line-calibrated metallicity of these galaxies, their overall distribution function, and ultimately use a sample of 200+ observations as spectral proxies for high-Z JWST EELGs. Organised and chaired the 4th-year physics interdisciplinary research group.

October 2024-December 2024 – Department of Physics, Durham University (BSc Equivalent)

‘A Comprehensive Photometric Model for the W-Type Binary GW Cep’

Supervisors: Prof Mark Swinbank, Prof Alastair Edge

Used Durham’s four 14-29inch telescopes to measure photometry and model the physical characteristics and long-term variability of a W-Uma-type contact binary. Data reduction and analysis including O-C, Lomb-Scargle and phase dispersion minimization, and semi-analytic techniques including openmpi-optimised simulated annealing and genetic evolution for the LTT equation. Results provided in the form of a 17-page report, including new discussions into the statistical viability of the system’s previously observed O’Connell effect and a discovery of a possible new third body for the system; **82% mark (highest in lab); won the Durham Florence Nightingale Award.**

September 2024-March 2025 – Department of Physics, Durham University

‘Investigating Colloidal Crystalline Freezing Using Massively Parallel Monte Carlo Simulations’

Supervisor: Dr. Nick Bristowe

Third-Year solid-state (soft matter) computational physics project. Build a python 26-thread massively-parallelized-Monte-Carlo simulation to perform fast random structure assimilations in 3D colloidal fluids, verifying experimental determinations of the coexistence pressure, density and freezing entropy. Assessment involved research poster, conference-style seminar and 9-page report (**79% mark – highest in group**).

SKILLS

Extensive background in computing; strong python skills. Experience working on HPC Clusters (Oxford Glamdring), GPU programming (including using `jax` and `torch`), and in low level languages C and Rust. Experienced in neural network architecture, including building latent-space-explainable unsupervised networks and supervised networks for statistical inference. Prior work on simulation-based- and Bayesian inference including solving posterior predictive distributions using neural inference.

Highly experienced in physical photometry and instrumentation. ~100 logged observation hours on the Durham Astrolab 14-29’ telescope domes for aperture timeseries photometry on binary star systems. Master’s project on IFU spectroscopy with imaging, including statistical inference on spectral properties using multiply lensed spectra (alongside work on JWST and HST photometry). Experience using MC and nested sampling for physical parameter inference on exoplanet timing detection (with binary stars); additional experience with binary star time domain modelling using PHOEBE and NIGHTFALL. Experience using radio cubes for HI source extraction using segmentation algorithms on (Oxford). Additionally experienced with LIGO/VIRGO data and signal processing from IoA summer project. Prior work with LLM deep learning, agentic tools and retrieval-augmented-generation.

OTHER RELEVANT EMPLOYMENT

2020-Present: Freelance – Orchestrator/Arranger/Music Publisher

Internationally respected orchestrator and music publisher. Orchestration services to three UK tours and the BBC along with consulting and editing services to the UK’s largest Brass Publisher. Work spans the UK and Northern Europe including headlining the London Palladium, Barbican and Ronnie Scott’s. Clients include GRAMMY-winning Cory Henry, the Metropole Orkest, and Danish Radio Symphony Orchestra. Commissioned by the Berliner Konzerthaus for work broadcast on DeutschlandRadio in summer 2024. Regular guest conductor for the National Youth Jazz Orchestra and have represented the UK Arts Council to the Royal Family. Winner of the 2025 international Au Arranging Prize.

2023-2024: Durham University – Levelling Up Physics

Educational mentor on Durham’s widening access physics higher education program. Provided academic-physics and pastoral support to many County Durham young people from underrepresented backgrounds.